Quadratic functions and technology

1

a
- i $(8, 7)$
- ii $(0, 263)$

b
- i $(-6, -9)$
- ii $(0, 135)$

c
- i $(-6, 4)$
- ii $(0, -140)$

d
- i $(2, -1)$
- ii $(0, -13)$

e
- i $\left(-\frac{1}{2}, \frac{33}{4}\right)$
- ii $(0, 9)$

f
- i $\left(-\frac{3}{8}, -\frac{25}{16}\right)$
- ii $(0, -1)$

2

a
- i $(0, 16)$
- ii No
- iii Increasing

b
- i $(0, -12)$
- ii No
- iii Decreasing

c
- i $(0, 125)$
- ii Yes, $5, -5$
- iii Decreasing

d
- i $(0, -64)$
- ii Yes, $4, -4$
- iii Increasing

3

a No

b Yes

c Yes

d No

4 $c = 25$

5

a
- i $-\frac{5}{2}$
- ii $\frac{25}{4}$
- iii $\left(-\frac{5}{2}, 0\right)$

b
- i $1, 5$
- ii -5
- iii $(3, 4)$

c
- i $1, 7$
- ii -7
- iii $(4, 9)$

6

a
- i $(-1, 3)$
- ii $(1, 3)$

b 2 units

7 Maximum value

8

a Minimum

b $(0, -4)$

9 $y = -\frac{81}{4}$

10

a -3.1

b -16.61

11

a

i $0, 6$ ii 0 iii 3 iv $(3, 9)$

b

i $0, -2$ ii 0 iii -1 iv $(-1, -1)$

c

i $0, -4$ ii 0 iii -2 iv $(-2, 4)$

d

i $-1, 3$ ii -3 iii 1 iv $(1, -4)$

e

i $-3, 1$ ii -3 iii -1 iv -4

f

i $2, 6$ ii -24 iii 4 iv 8

12

a 2.56

b 4.45

c

i Yes ii No iii No

d $5.12, 0.00$

13

a 2.75

b -9.10

c

i No ii Yes iii No

d $-0.42, 5.91$

14

a 1.55

b 2.95

c

i No ii Yes iii No iv No

d $3.58, -0.47$

15

a -1.38

b 0.84

c

i No ii No iii No iv Yes

d $-0.69, -2.08$

16 Move the graph downwards by 5 units.

17 Move the graph to the right by 3 units.

18 Narrows the graph

19

a $y=x^2+2$ b $y=-x^2$ c $y=-x^2-2$ d $y=(x+2)^2+2$

20 $y=(x-2)^2+3$

21

a $x=0, x=\frac{8}{7}$ b $x=-\frac{1}{2}, x=3$ c $x=\frac{3}{2}, x=-1$ d $x=-\frac{1}{4}, x=5$

e $x=-1$ f $x=-4, x=-3$

22

a 1 b 2 c 0

23 $2.000, 1.667$

24 $1.584, -0.859$

Additional questions

25

a $(0,0)$ b The initial distance fallen by the object

26

a $0, 30$

b 15

c The largest rectangular area that can be fenced off with $60\ \mathrm{m}$ of wire.

27 $\pm 14\ \mathrm{m/s}$

28

a $176.4\ \mathrm{m}$ b $6\ \mathrm{s}$ c $12\ \mathrm{s}$

29

a $900 \mathrm{~m}$

b Initially, the object has not fallen any distance.

c $13.55 \mathrm{~s}$

d After $13.55 \mathrm{~s}$ the object is on the ground.

30

a $4 \mathrm{~s}$

b $10 \mathrm{~s}$